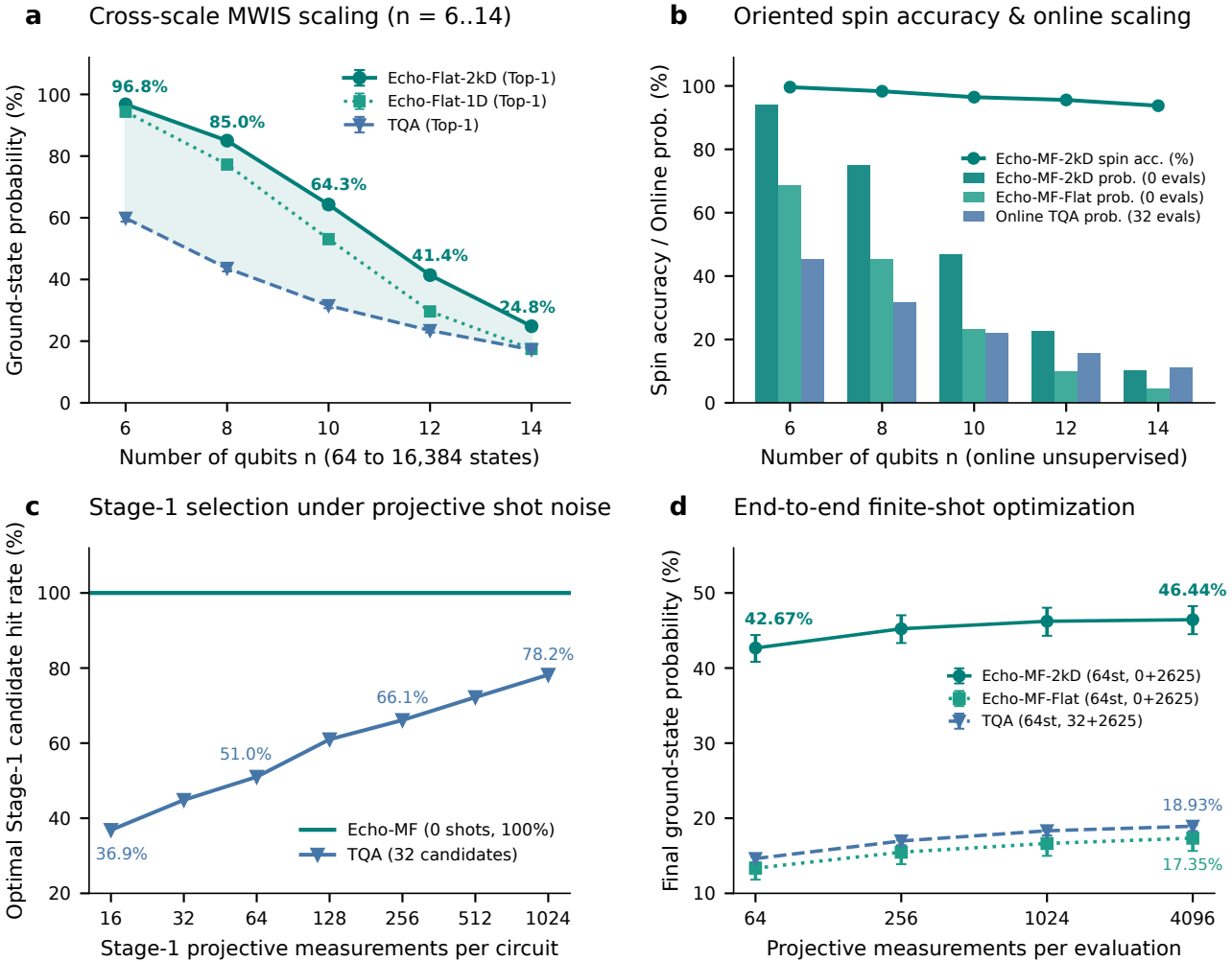
a
Cross-scale MWIS scaling (n = 6..14)
Ground-state probability (%)
Number of qubits n (64 to 16,384 states)
Echo-Flat-2kD (Top-1)
Echo-Flat-1D (Top-1)
TQA (Top-1)
96.8%
85.0%
64.3%
41.4%
24.8%
b
Oriented spin accuracy & online scaling
Spin accuracy / Online prob. (%)
Number of qubits n (online unsupervised)
Echo-MF-2kD spin acc. (%)
Echo-MF-2kD prob. (0 evals)
Echo-MF-Flat prob. (0 evals)
Online TQA prob. (32 evals)
c
Stage-1 selection under projective shot noise
Optimal Stage-1 candidate hit rate (%)
Stage-1 projective measurements per circuit
Echo-MF (0 shots, 100%)
TQA (32 candidates)
36.9%
51.0%
66.1%
78.2%
d
End-to-end finite-shot optimization
Final ground-state probability (%)
Projective measurements per evaluation
Echo-MF-2kD (64st, 0+2625)
Echo-MF-Flat (64st, 0+2625)
TQA (64st, 32+2625)
42.67%
46.44%
18.93%
17.35%